Exercise 1

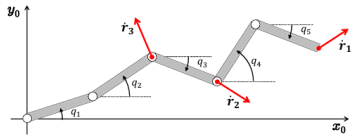


Figure 1: The 5R planar robot with three desired task velocities $\dot{r}_1$, $\dot{r}_2$ and $\dot{r}_3$.

Consider a 5R planar robot with links of unitary length in a generic configuration q, where q_i, $i = 1, \dots, 5$, are absolute angles with respect to the x_0 axis, as shown in Fig. 1. The robot is controlled by the absolute joint velocity $\dot{q} \in \mathbb{R}^5$. Three velocity tasks $\dot{r}_i \in \mathbb{R}^2$, $i = 1, 2, 3$, are assigned with priority, respectively, the first for the end-effector, the second for the tip of link 3, and the third for the tip of link 2.

i) Write down the computations to be performed in order to determine the joint velocity command $\dot{q}_{TP}$ according to the Task Priority (TP) method.

ii) Compute the numerical value of $\dot{q}_{TP}$ for the following data:

$$q = (0, \pi/3, -\pi/4, \pi/2, -\pi/3) \ \text{[rad]} \qquad \dot{r}_1 = (2, 3) \qquad \dot{r}_2 = (2, -0.5) \qquad \dot{r}_3 = (-1, 0) \ \text{[m/s]}.$$

Are the tasks exactly realized? If not, compute the norm of the velocity error for each of them.

iii) With the same data as in *ii)*, compute the joint velocity command $\dot{q}_{TA}$ according to the Task Augmentation (TA) method, i.e., without any priority. Determine the norm of the velocity error for each task that is not exactly realized. Discuss the obtained TA results in comparison with the TP method.

iv) Keeping the same task velocities $\dot{r}_1$ and $\dot{r}_2$ as in *ii)*, determine, if at all possible, a task velocity $\dot{r}_3 \in \mathbb{R}^2$ such that the joint velocity command $\dot{q}_{TA}$ obtained with the TA method realizes exactly all three tasks.

v) For the solution in *ii)*, find the corresponding Denavit–Hartenberg (relative) joint velocity $\dot{\theta}_{TP}$.

The recursive formula is $\dot{q}_K = \dot{q}_{K-1} + (J_K P_{A,K-1})^{\#}(\dot{r}_K - J_K \dot{q}_{K-1})$ where $P_{A,i}$ is the null space projector $P_{A,K} = P_{A,K-1} - (J_K P_{A,K-1})^{\#} J_K P_{A,K-1}$

First of all, i write the DK for the 3 points:

$$r_3 = \begin{pmatrix} c_1 + c_2 \\ s_1 + s_2 \end{pmatrix} \qquad r_2 = \begin{pmatrix} c_1 + c_2 + c_3 \\ s_1 + s_2 + s_3 \end{pmatrix} \qquad r_1 = \sum_{i=1}^{5} \begin{pmatrix} c_i \\ s_i \end{pmatrix}$$

priority: r_1, r_2, r_3

Now i compute the Jacobians

$$J_3 = \begin{pmatrix} -s_1 & -s_2 & 0 & 0 & 0 \\ c_1 & c_2 & 0 & 0 & 0 \end{pmatrix} \qquad J_2 = \begin{pmatrix} -s_1 & -s_2 & -s_3 & 0 & 0 \\ c_1 & c_2 & c_3 & 0 & 0 \end{pmatrix} \qquad J_1 = \begin{pmatrix} -s_1 & -s_2 & -s_3 & -s_4 & -s_5 \\ c_1 & c_2 & c_3 & c_4 & c_5 \end{pmatrix}$$

Now i compute the numerical values with $q = \begin{pmatrix} 0 & \frac{\pi}{3} & -\frac{\pi}{4} & \frac{\pi}{2} & -\frac{\pi}{3} \end{pmatrix}^T$

$$J_1 = \begin{pmatrix} 0 & -\sqrt{3}/2 & \sqrt{2}/2 & -1 & \sqrt{3}/2 \\ 1 & 1/2 & \sqrt{2}/2 & 0 & 1/2 \end{pmatrix} \quad J_2 = \begin{pmatrix} 0 & -\sqrt{3}/2 & \sqrt{2}/2 & 0 & 0 \\ 1 & 1/2 & \sqrt{2}/2 & 0 & 0 \end{pmatrix} \quad J_3 = \begin{pmatrix} 0 & -\sqrt{3}/2 & 0 & 0 & 0 \\ 1 & 1/2 & 0 & 0 & 0 \end{pmatrix}$$

I compute $\dot{q}_3 = \dot{q}_{TP}$ with the following script:

```
J1=[0,-sqrt(3)/2, sqrt(2)/2,-1,sqrt(3)/2;
    1,1/2, sqrt(2)/2,0,1/2];

J2=[0,-sqrt(3)/2, sqrt(2)/2,0,0;
    1,1/2, sqrt(2)/2,0,0];

J3=[0,-sqrt(3)/2, 0,0,0;
    1,1/2, 0,0,0];

dr1=[2;3]
dr2=[2;-0.5]
dr3=[-1;0]
```

```
J1_plus = pinv(J1);

P0=eye(5)
P1 = P0- J1_plus*J1;

dq1=pinv(J1*P0)*(dr1)

P2 = P1-pinv(J2*P1)*(J2*P1)

dq2 = dq1+pinv(J2*P1)*(dr2-J2*dq1)

dq3 = dq2+pinv(J3*P2)*(dr3-J3*dq2)
```

$\Rightarrow$ provides $\dot{q}_{TP} = \begin{pmatrix} -1.317 & -0.866 & 1.7678 & 6.0622 & 7 \end{pmatrix}^T$

Let's compute the error:

$$e = J_A \dot{q}_{TP} - \begin{pmatrix} \dot{r}_1 \\ \dot{r}_2 \\ \dot{r}_3 \end{pmatrix} = \begin{bmatrix} 0 & -\sqrt{3}/2 & \sqrt{2}/2 & -1 & \sqrt{3}/2 \\ 1 & 1/2 & \sqrt{2}/2 & 0 & 1/2 \\ 0 & -\sqrt{3}/2 & \sqrt{2}/2 & 0 & 0 \\ 1 & 1/2 & \sqrt{2}/2 & 0 & 0 \\ 0 & -\sqrt{3}/2 & 0 & 0 & 0 \\ 1 & 1/2 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1.317 \\ -0.866 \\ 1.7678 \\ 6.0622 \\ 7 \end{bmatrix} - \begin{pmatrix} \dot{r}_1 \\ \dot{r}_2 \\ \dot{r}_3 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1.75 \\ -1.75 \end{bmatrix} \Rightarrow \|e\| = 2.4743$$

Exercise 2

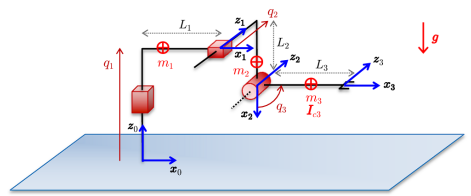


Figure 2: A PPR spatial robot with its D–H frames and relevant kinematic and dynamic parameters.

Figure 2 shows a 3-dof robot moving under gravity in the 3D space, with the associated Denavit–Hartenberg (D–H) frames. The first two joints are prismatic and the third is revolute. The most relevant kinematic and dynamic parameters are also shown. The center of mass of each link is assumed to be on the link kinematic axis ($\boldsymbol{x}_i$ for link i). No special assumption holds for the barycentric inertia matrix $\boldsymbol{I}_{c3}$ of link 3.

i) Derive all terms of the dynamic model in the Lagrangian form

$$\boldsymbol{M}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q},\dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}.$$

ii) Find two different factorization matrices $\boldsymbol{S}_i(\boldsymbol{q},\dot{\boldsymbol{q}})$, $i = 1,2$, such that $\boldsymbol{S}_i(\boldsymbol{q},\dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}} = \boldsymbol{c}(\boldsymbol{q},\dot{\boldsymbol{q}})$; show that matrix $\boldsymbol{M} - 2\boldsymbol{S}$ is skew-symmetric when $\boldsymbol{S} = \boldsymbol{S}_1$, whereas it is not when $\boldsymbol{S} = \boldsymbol{S}_2$.

iii) Determine a minimal linear parametrization of the model useful for system identification

$$\boldsymbol{Y}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}})\,\boldsymbol{a} = \boldsymbol{\tau},$$

providing the symbolic expression of the (unknown) dynamic coefficients $\boldsymbol{a} \in \mathbb{R}^p$ and of the $3 \times p$ regressor matrix $\boldsymbol{Y}$. Assume that the acceleration of gravity $g_0 = 9.81$ m/s^2 is known.

iv) Give the expression of the torque $\boldsymbol{\tau}_d(t) = (\tau_{d1}(t), \tau_{d2}(t)), \tau_{d3}(t))$ needed to execute the desired motion $\boldsymbol{q}_d(t) = (0, \sin(2\omega t), \omega t)$, for all $t \geq 0$. Which should be the initial state $(\boldsymbol{q}(0), \dot{\boldsymbol{q}}(0))$ of the robot so that this joint trajectory is perfectly executed from $t = 0$ on? Discuss the physics of the results.

Without having to define the DH matrices, is easy to see that the position of each center of mass in Frame 0 is:

$$^0P_{c1} = \begin{pmatrix} d_{c1} \\ 0 \\ q_1 \end{pmatrix} \qquad ^0P_{c2} = \begin{pmatrix} L_1 \\ q_2 \\ q_1 - d_{c2} \end{pmatrix} \qquad ^0P_{c3} = \begin{pmatrix} L_1 + d_{c3}\cos q_3 \\ q_2 \\ q_1 - L_2 + d_{c3}\sin q_3 \end{pmatrix}$$

these quantities are useful to compute the potential energy. For the kinetic instead:

$$^0v_{c1} = (0 \quad 0 \quad \dot{q}_1)^T \qquad ^0v_{c2} = (0 \quad \dot{q}_2 \quad \dot{q}_1)^T \qquad ^0v_{c3} = (-d_{c3}c_3\dot{q}_3 \quad \dot{q}_2 \quad \dot{q}_1 + d_{c3}s_3\dot{q}_3)^T$$

$$^0\omega_1 = 0 \qquad ^0\omega_2 = 0 \qquad ^3\omega_3 = (0 \quad 0 \quad -\dot{q}_3)$$

$$\Rightarrow T_1 = \frac{1}{2}m_1\dot{q}_1^2 \qquad T_2 = \frac{1}{2}m_2\dot{q}_1^2 + \frac{1}{2}m_2\dot{q}_2^2$$

$$T_3 = \frac{1}{2}m_3\left(d_{c3}^2 s_3^2\dot{q}_3^2 + \dot{q}_2^2 + \dot{q}_1^2 + d_{c3}^2c_3^2\dot{q}_3^2 + 2d_{c3}s_3\dot{q}_1\dot{q}_3\right) + \frac{1}{2}(0 \quad 0 \quad -\dot{q}_3)\, I_{c3}\begin{pmatrix} 0 \\ 0 \\ -\dot{q}_3 \end{pmatrix} =$$

$$\frac{1}{2}m_3\left[\dot{q}_3^2 d_{c3}^2 + \dot{q}_2^2 + \dot{q}_1^2 + \dot{q}_1\dot{q}_3(2d_{c3}s_3)\right] + \frac{1}{2}I_{c3zz}\dot{q}_3^2 \Rightarrow$$

$$T = \frac{1}{2}\dot{q}_1^2(m_1 + m_2 + m_3) + \frac{1}{2}\dot{q}_2^2(m_2 + m_3) + \frac{1}{2}\dot{q}_3^2(m_3 d_{c3}^2 + I_{c3zz}) + \frac{1}{2}\dot{q}_1\dot{q}_3(2d_{c3}s_3)$$

$$\Rightarrow M = \begin{bmatrix} m_1 + m_2 + m_3 & 0 & m_3 d_{c3} s_3 \\ 0 & m_2 + m_3 & 0 \\ d_{c3}s_3 m_3 & 0 & m_3 d_{c3}^2 + I_{c3zz} \end{bmatrix} = \begin{pmatrix} \rho_1 & 0 & \rho_2 s_3 \\ 0 & \rho_3 & 0 \\ \rho_2 s_3 & 0 & \rho_4 \end{pmatrix}$$

$$\rho_1 = \sum_i m_i \qquad \rho_2 = d_{c3}m_3 \qquad \rho_4 = m_3 d_{c3}^2 + I_{c3zz} \qquad \rho_3 = m_2 + m_3$$

Now i compute the potential energy using $^0P_{ci}$:

$$U = -g_0\left(m_1 q_1 + m_2(q_1 - d_{c2}) + m_3(q_1 - L_2 + d_{c3}s_3)\right)$$

$$\frac{\partial U}{\partial q_1} = -g_0(m_1 + m_2 + m_3) = \rho_5 \qquad \frac{\partial U}{\partial q_2} = 0 \qquad \frac{\partial U}{\partial q_3} = \underbrace{-g_0 m_3 d_{c3}}_{\rho_6} c_3 = \rho_6 c_3$$

Now i have to compute the centrifugal and Coridis vector $c(q,\dot{q})$. Given $M(q)$, i consider $C_i = \frac{1}{2}\left(\frac{\partial M_i}{\partial q} + \left(\frac{\partial M_i}{\partial q}\right)^T - \frac{\partial M}{\partial q_i}\right)$

$$C_1 = \frac{1}{2}\left(\frac{\partial M_1}{\partial q} + \left(\frac{\partial M_1}{\partial q}\right)^T - \overset{0}{\frac{\partial M}{\partial q_1}}\right) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \rho_2 c_3 \end{pmatrix}$$

$$C_2 = \frac{1}{2}\left(\frac{\partial M_2}{\partial q} + \left(\frac{\partial M_2}{\partial q}\right)^T - \overset{0}{\frac{\partial M}{\partial q_2}}\right) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$C_3 = \frac{1}{2}\left(\frac{\partial M_3}{\partial q} + \left(\frac{\partial M_3}{\partial q}\right)^T - \frac{\partial M}{\partial q_3}\right) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Since $c_i = \dot{q}^T C_i \dot{q} \Rightarrow c = (\rho_2 c_3 \dot{q}_3^2, 0, 0)^T$

The complete model is

$$\begin{bmatrix} \rho_1 & 0 & \rho_2 s_3 \\ 0 & \rho_3 & 0 \\ \rho_2 s_3 & 0 & \rho_4 \end{bmatrix} \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \\ \ddot{q}_3 \end{bmatrix} + \begin{bmatrix} \rho_2 c_3 \dot{q}_3^2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \rho_5 \\ 0 \\ \rho_6 c_3 \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix}$$

since $c = (\rho_2 c_3 \dot{q}_3^2, 0, 0)^T \Rightarrow$

$$S = \begin{pmatrix} 0 & 0 & \rho_2 c_3 \dot{q}_3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow c = S\dot{q}$$

let's check if $\dot{M} - 2S$ is skew symmetric

$$\dot{M} - 2S = \begin{pmatrix} 0 & 0 & \rho_2 c_3 \dot{q}_3 \\ 0 & 0 & 0 \\ \rho_2 c_3 \dot{q}_3 & 0 & 0 \end{pmatrix} - 2\begin{pmatrix} 0 & 0 & \rho_2 c_3 \dot{q}_3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -\rho_2 c_3 \dot{q}_3 \\ 0 & 0 & 0 \\ \rho_2 c_3 \dot{q}_3 & 0 & 0 \end{pmatrix}$$

is skew symmetric

Another Factorization s.t. $\dot{M} - 2S$ is not skew symmetric is:

$$S' = \begin{pmatrix} 0 & 0 & \rho_2 c_3 \dot{q}_3 \\ 0 & 0 & 0 \\ -\dot{q}_3 & 0 & \dot{q}_1 \end{pmatrix} \Rightarrow \dot{M} - 2S' = \begin{pmatrix} 0 & 0 & \rho_2 c_3 \dot{q}_3 \\ 0 & 0 & 0 \\ \rho_2 c_3 \dot{q}_3 & 0 & 0 \end{pmatrix} - 2\begin{pmatrix} 0 & 0 & \rho_2 c_3 \dot{q}_3 \\ 0 & 0 & 0 \\ -\dot{q}_3 & 0 & \dot{q}_1 \end{pmatrix} =$$

$$\begin{pmatrix} 0 & 0 & -\rho_2 c_3 \dot{q}_3 \\ 0 & 0 & 0 \\ \rho_2 c_3 \dot{q}_3 - 2\dot{q}_3 & 0 & 2\dot{q}_1 \end{pmatrix}$$

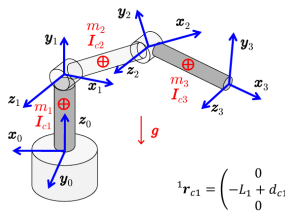
the desired trajectory is : $\dot{q}_d = (0,\ 2\omega\cos(\omega t),\ \omega)^T$ $\quad \ddot{q}_d = (0,\ -4\omega^2\sin(\omega t),\ 0)^T$

$$\Rightarrow \begin{bmatrix} \rho_1 & 0 & \rho_2 s_3 \\ 0 & \rho_3 & 0 \\ \rho_2 s_3 & 0 & \rho_4 \end{bmatrix} \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \\ \ddot{q}_3 \end{bmatrix} + \begin{bmatrix} \rho_2 c_3 \dot{q}_3^2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \rho_5 \\ 0 \\ \rho_6 c_3 \end{bmatrix}$$

; substitute $q_d, \dot{q}_d, \ddot{q}_d$

$$\Rightarrow \tau_d = \begin{cases} \rho_5 + \rho_2 \cos(\omega t)\,\omega^2 \\ -4\omega^2 \sin(\omega t)\,\rho_3 \\ \rho_3 \cos(\omega t) \end{cases}$$

Exercise 3



i	α_i	a_i	d_i	θ_i
1	$\pi/2$	0	L_1	q_1
2	0	L_2	0	q_2
3	0	L_3	0	q_3

$${}^{i}I_{ci} = \begin{pmatrix} I_{xxi} & 0 & 0 \\ 0 & I_{yyi} & 0 \\ 0 & 0 & I_{zzi} \end{pmatrix}, \qquad i = 1,2,3$$

$${}^{1}r_{c1} = \begin{pmatrix} 0 \\ -L_1 + d_{c1} \\ 0 \end{pmatrix} \qquad {}^{i}r_{ci} = \begin{pmatrix} -L_i + d_{ci} \\ 0 \\ 0 \end{pmatrix}, \qquad i = 2,3$$

Figure 3: A 3R robot with D–H frames and table of parameters, and the dynamic assumptions.

For a 3R elbow-type robot, all the kinematic and dynamic information is given in Fig. 3. Derive the robot kinetic energy $T(\boldsymbol{q}, \dot{\boldsymbol{q}})$ using the recursive algorithm with moving frames. At a given robot state $(\boldsymbol{q}, \dot{\boldsymbol{q}})$, consider then the dynamic optimization problem

$$\min \frac{1}{2}(\boldsymbol{\tau} - \boldsymbol{n}(\boldsymbol{q}, \dot{\boldsymbol{q}}))^T(\boldsymbol{\tau} - \boldsymbol{n}(\boldsymbol{q}, \dot{\boldsymbol{q}})) \qquad \text{s.t.} \quad \begin{cases} \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{n}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \boldsymbol{\tau} \\ \boldsymbol{J}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \dot{\boldsymbol{J}}(\boldsymbol{q})\dot{\boldsymbol{q}} = \ddot{\boldsymbol{r}}_d, \end{cases}$$

where $\boldsymbol{J}(\boldsymbol{q})$ is the 2×3 Jacobian associated to the task $\boldsymbol{r} = (p_x, p_y) = \boldsymbol{f}(\boldsymbol{q})$, i.e., two of the three coordinates of the robot end-effector position $\boldsymbol{p}$. Provide the symbolic expression of the joint acceleration $\ddot{\boldsymbol{q}}$ that solves this problem. Evaluate numerically this solution when $\boldsymbol{q} = (3\pi/2, \pi/4, 0)$ rad], $\dot{\boldsymbol{q}} = \boldsymbol{0}$, and $\ddot{\boldsymbol{r}}_d = (-1, -1)$ [m/s^2] and with the data

$$L_i = 0.5 \quad d_{ci} = 0.5\,L_i \quad [\text{m}] \qquad m_i = 2 \text{ kg} \qquad \text{for } i = 1, 2, 3,$$

$$I_{yy1} = \tfrac{1}{40}\,m_1 \qquad I_{xxi} = \tfrac{1}{40}\,m_i \quad I_{yyi} = I_{zzi} = \tfrac{1}{12}\,m_i\,(0.05 + L_i^2) \quad [\text{kgm}^2] \qquad \text{for } i = 2, 3.$$

I write the three DH transformations.

$${}^0A_1 = \begin{pmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & L_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^1A_2 = \begin{pmatrix} c_2 & -s_2 & 0 & L_2 c_2 \\ s_2 & c_2 & 0 & L_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^2A_3 = \begin{pmatrix} c_3 & -s_3 & 0 & L_3 c_3 \\ s_3 & c_3 & 0 & L_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Now ; apply the recursive algorithm:

$i=1)$ $${}^1\omega_1 = {}^0R_1^T\left[0 + (0\ \ 0\ \ \dot{q}_1)^T\right] = \begin{pmatrix} c_1 & s_1 & 0 \\ 0 & 0 & 1 \\ s_1 & -c_1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ \dot{q}_1 \\ 0 \end{pmatrix}$$

$${}^1v_1 = {}^0R_1^T\left[0 + \begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ L_1 \end{pmatrix}\right] = 0 \Rightarrow v_{c1} = 0 + \begin{pmatrix} 0 \\ \dot{q}_1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -L_1 + d_{c1} \\ 0 \end{pmatrix} = 0$$

$i=2)$ $${}^2\omega_2 = {}^1R_2^T\left[(0\ \ \dot{q}_1\ \ 0)^T + (0\ \ 0\ \ \dot{q}_2)^T\right] = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ \dot{q}_1 \\ \dot{q}_2 \end{pmatrix} = \begin{pmatrix} \dot{q}_1 s_2 \\ \dot{q}_1 c_2 \\ \dot{q}_2 \end{pmatrix}$$

$${}^2v_2 = {}^1R_2^T\left[0 + \begin{pmatrix} 0 \\ \dot{q}_1 \\ \dot{q}_2 \end{pmatrix} \times \begin{pmatrix} L_2 c_2 \\ L_2 s_2 \\ 0 \end{pmatrix}\right] = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -\dot{q}_2 & \dot{q}_1 \\ \dot{q}_2 & 0 & 0 \\ -\dot{q}_1 & 0 & 0 \end{pmatrix}\begin{pmatrix} L_2 c_2 \\ L_2 s_2 \\ 0 \end{pmatrix} = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} -L_2 \dot{q}_2 s_2 \\ L_2 \dot{q}_2 c_2 \\ -L_2 \dot{q}_1 c_2 \end{pmatrix}$$

$$= \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} -L_2 \dot{q}_2 s_2 \\ L_2 \dot{q}_2 c_2 \\ -L_2 \dot{q}_1 c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ L_2 \dot{q}_2 \\ -L_2 \dot{q}_1 c_2 \end{pmatrix} \Rightarrow v_{c2} = \begin{pmatrix} 0 \\ L_2 \dot{q}_2 \\ -L_2 \dot{q}_1 c_2 \end{pmatrix} + \begin{pmatrix} \dot{q}_1 s_2 \\ \dot{q}_1 c_2 \\ \dot{q}_2 \end{pmatrix} \times \begin{pmatrix} -L_2 + d_{c2} \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \dot{q}_2 d_{c2} \\ -d_{c2} \dot{q}_1 c_2 \end{pmatrix}$$

$i = 3$)

$$ {}^3\omega_3 = {}^2R_3^T\left[{}^2\omega_2 + \begin{pmatrix}0 & 0 & \dot q_3\end{pmatrix}^T\right] = \begin{pmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}\dot q_1 s_2 \\ \dot q_1 c_2 \\ \dot q_2 + \dot q_3\end{pmatrix} = \begin{pmatrix}\dot q_1 s_2 c_3 + \dot q_1 c_2 s_3 \\ -\dot q_1 s_2 s_3 + \dot q_1 c_2 c_3 \\ \dot q_2 + \dot q_3\end{pmatrix} = \begin{pmatrix}\dot q_1 s_{23} \\ \dot q_1 c_{23} \\ \dot q_2 + \dot q_3\end{pmatrix} $$

$$ {}^3v_3 = {}^2R_3^T\left[{}^2v_2 + \begin{pmatrix}\dot q_1 s_2 \\ \dot q_1 c_2 \\ \dot q_2 + \dot q_3\end{pmatrix} \times \begin{pmatrix} L_3 c_3 \\ L_3 s_3 \\ 0\end{pmatrix}\right] = {}^2R_3^T\left[{}^2v_2 + \begin{pmatrix} 0 & -(\dot q_2 + \dot q_3) & \dot q_1 c_2 \\ \dot q_2 + \dot q_3 & 0 & -\dot q_1 s_2 \\ -\dot q_1 c_2 & \dot q_1 s_2 & 0\end{pmatrix}\begin{pmatrix} L_3 c_3 \\ L_3 s_3 \\ 0\end{pmatrix}\right] $$

$$ = \begin{pmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1\end{pmatrix}\left(\begin{pmatrix} 0 \\ L_2 \dot q_2 \\ -L_2 \dot q_1 c_2\end{pmatrix} + \begin{pmatrix} 0 & -(\dot q_2 + \dot q_3) & \dot q_1 c_2 \\ \dot q_2 + \dot q_3 & 0 & -\dot q_1 s_2 \\ -\dot q_1 c_2 & \dot q_1 s_2 & 0\end{pmatrix}\begin{pmatrix} L_3 c_3 \\ L_3 s_3 \\ 0\end{pmatrix}\right) = \begin{pmatrix} L_2 \dot q_2 s_3 \\ L_3(\dot q_2 + \dot q_3) + L_2 \dot q_2 c_3 \\ -\dot q_1 (L_3 c_{23} + L_2 c_2)\end{pmatrix} $$

$$ v_{c_3} = \begin{pmatrix} L_2 \dot q_2 s_3 \\ L_3(\dot q_2 + \dot q_3) + L_2 \dot q_2 c_3 \\ -\dot q_1 (L_3 c_{23} + L_2 c_2)\end{pmatrix} + \begin{pmatrix}\dot q_1 s_{23} \\ \dot q_1 c_{23} \\ \dot q_2 + \dot q_3\end{pmatrix} \times \begin{pmatrix} -L_3 + d_{c3} \\ 0 \\ 0\end{pmatrix} = \begin{pmatrix} L_2 \dot q_2 s_3 \\ L_2 \dot q_2 c_3 + d_{c_3}(\dot q_2 + \dot q_3) \\ -\dot q_1 (L_2 c_2 + d_{c3} c_{23})\end{pmatrix} $$

At this point i compute :

$$ \|v_{c_1}\|^2 = 0 \qquad \|v_{c_2}\|^2 = \dot q_2^2 d_{c_2}^2 + \dot q_1^2 d_{c_2}^2 c_2^2 $$

$$ \|v_{c_3}\|^2 = \dot q_2^2 L_2^2 s_3^2 + \dot q_2^2 L_2^2 c_3^2 + d_{c3}^2(\dot q_2 + \dot q_3)^2 + 2 L_2 c_3 d_{c3}(\dot q_2^2 + \dot q_2 \dot q_3) + \dot q_1^2 (L_2^2 c_2^2 + d_{c3}^2 c_{23}^2 + 2 L_2 d_{c3} c_2 c_{23}) $$

$$ \omega_1^T I_{c_1} \omega_1 = I_{yy_1} \dot q_1^2 \qquad \omega_2^T I_{c_2} \omega_2 = I_{2xx} \dot q_1^2 s_2^2 + I_{2yy} \dot q_1^2 c_2^2 + I_{2zz} \dot q_2^2 $$

$$ \omega_3^T I_{c_3} \omega_3 = I_{3xx} \dot q_1^2 s_{23}^2 + I_{3yy} \dot q_1^2 c_{23}^2 + I_{3zz}(\dot q_2^2 + \dot q_3^3 + 2 \dot q_2 \dot q_3) $$

So :

$$ T = \frac{1}{2}\dot q_1^2 \left[m_2 d_{c_2}^2 c_2^2 + m_3(L_2^2 c_2^2 + d_{c3}^2 c_{23}^2 + 2 L_2 d_{c3} c_2 c_{23}) + I_{yy_1} + I_{xx_2} s_2^2 + I_{yy_2} c_2^2 + I_{xx_3} s_{23}^2 + I_{yy_3} c_{23}^2\right] $$

$$ + \frac{1}{2}\dot q_2^2 \left[m_2 d_{c_2}^2 + m_3(L_2^2 + d_{c3}^2 + 2 L_2 c_3 d_{c3}) + I_{2zz} + I_{3zz}\right] $$

$$ + \frac{1}{2}\dot q_3 \left[m_3 d_{c3}^2 + I_{zz3}\right] $$

$$ + \frac{1}{2}\dot q_2 \dot q_3 2 \left[d_{c3}^2 + L_2 c_3 d_{c3} + I_{zz3}\right] $$

$\Rightarrow$ the matrix M is defined by this Factorization.

$$ M_{11} = m_2 d_{c_2}^2 c_2^2 + m_3(L_2^2 c_2^2 + d_{c3}^2 c_{23}^2 + 2 L_2 d_{c3} c_2 c_{23}) + I_{yy_1} + I_{xx_2} s_2^2 + I_{yy_2} c_2^2 + I_{xx_3} s_{23}^2 + I_{yy_3} c_{23}^2 $$

$$ M_{22} = m_2 d_{c_2}^2 + m_3(L_2^2 + d_{c3}^2 + 2 L_2 c_3 d_{c3}) + I_{2zz} + I_{3zz} $$

$$ M_{33} \; m_3 d_{c3}^2 + I_{zz3} \qquad M_{13} = M_{12} = 0 $$

$$ M_{23} = M_{32} = d_{c3}^2 + L_2 c_3 d_{c3} + I_{zz3} $$

When $(q, \dot{q})$ are fixed, there is a 1-by-1 mapping between $\ddot{q}$ and τ

$M\ddot{q} + n = \tau \Rightarrow \ddot{q} = M^{-1}(\tau - n)$ we know that M is non singular.

Our joint acceleration should satisfy

$J\ddot{q} + \dot{J}\dot{q} = \ddot{r}_d \Rightarrow J\ddot{q} = \ddot{r}_d - \dot{J}\dot{q}$. The objective function can be rewritten

$\min \frac{1}{2}(\tau - n)^T(\tau - n) = \frac{1}{2}(M\ddot{q})^T M q = \frac{1}{2}\ddot{q}^T M^T M \ddot{q}$ so we are trying to minimized the weighted norm. The solution is of course given by:

$$J_W^{\#} = W^{-1} J^T (J W^{-1} J^T)^{-1} = (M^T M)^{-1}(J (M^T M)^{-1} J^T)^{-1} \Rightarrow \ddot{q}_d = J_W^{\#}(\ddot{r}_d - \dot{J}\dot{q})$$

$W = M^T M$

with the numerical data:

$M_{11} = 10 \cdot 0.5^2 \cos(\frac{\pi}{4})^2 + \frac{1}{40} + \frac{1}{20}\sin(\frac{\pi}{4})^2 + \frac{1}{6}\cos(\frac{\pi}{4})^2 = 1.3833$

$M_{22} = 2.6666$

$M_{33} = 0.5833 = M_{23}$

$$\Rightarrow M = \begin{pmatrix} 1.3833 & 0 & 0 \\ 0 & 2.6666 & 0.5833 \\ 0 & 0.5833 & 0.5833 \end{pmatrix} \Rightarrow (M^T M)^{-1} = \begin{pmatrix} 0.5225 & 0 & 0 \\ 0 & 0.4608 & -1.283 \\ 0 & -1.283 & 5.0457 \end{pmatrix}$$

The DK is

$$P_e = \begin{cases} c_1(L_3 c_{23} + L_2 c_2) \\ s_1(L_3 c_{23} + L_2 c_2) \end{cases} \Rightarrow J = \begin{pmatrix} -s_1(L_3 c_{23} + L_2 c_2) & -c_1(L_3 s_{23} - L_2 s_2) & -c_1 L_3 s_{23} \\ c_1(L_3 c_{23} + L_2 c_2) & -s_1(L_3 s_{23} - L_2 s_2) & -s_1 L_3 s_{23} \end{pmatrix}$$

$$\Rightarrow \text{evaluated}: J = \begin{pmatrix} \sqrt{2}/2 & 0 & 0 \\ 0 & 0 & \sqrt{2}/4 \end{pmatrix} \Rightarrow J_W^{\#}$$

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